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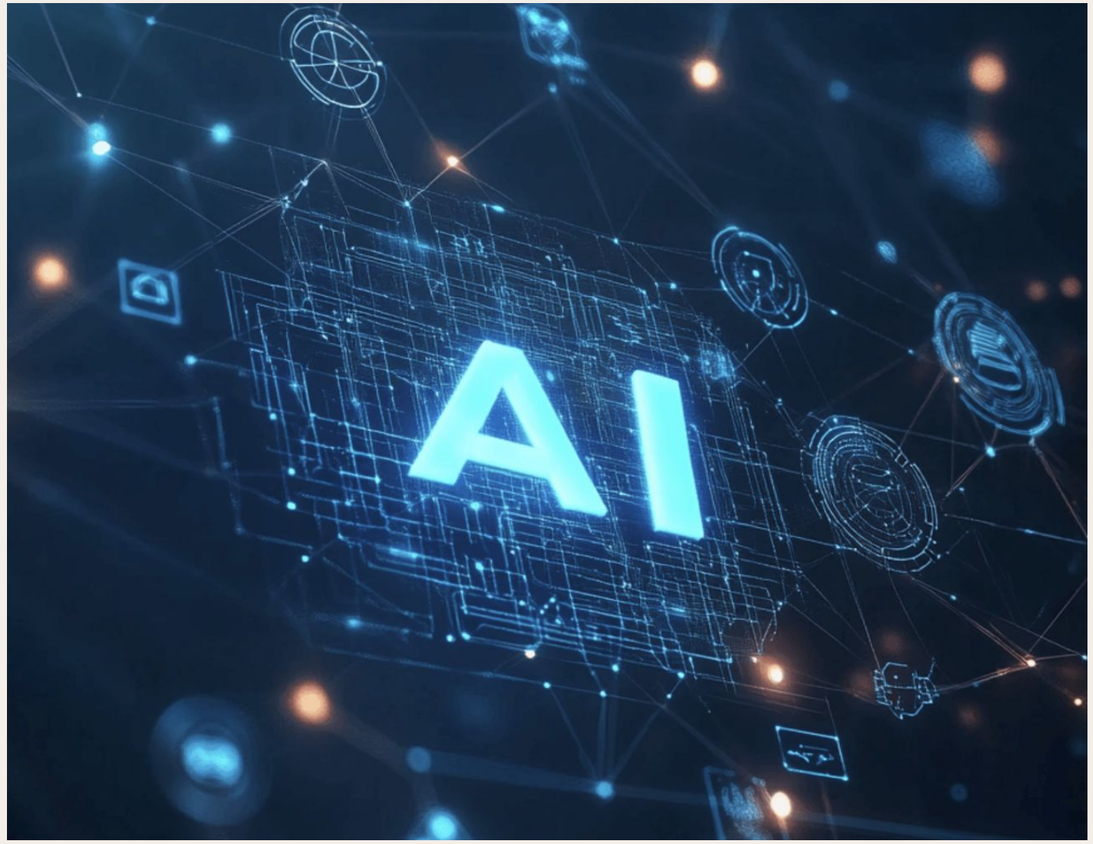
A Theory-of-Mind Driven Error Taxonomy for Deception Detection in Courtroom Testimonies

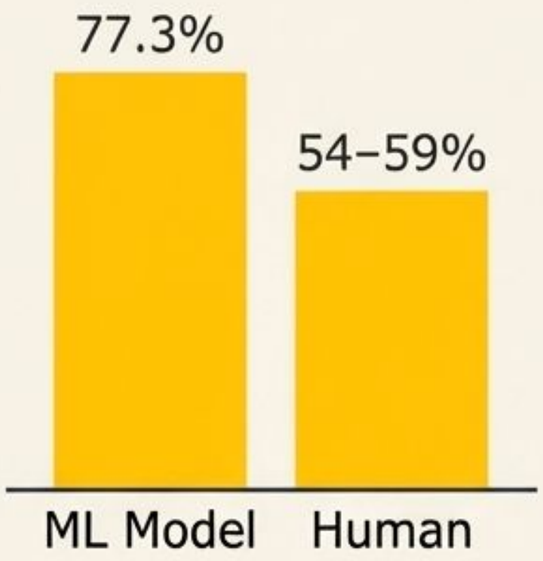
Yayun Zheng

MA Psychology Student New York University

BELMONT UNIVERSITY


AI

The Black Box Problem in Forensic AI



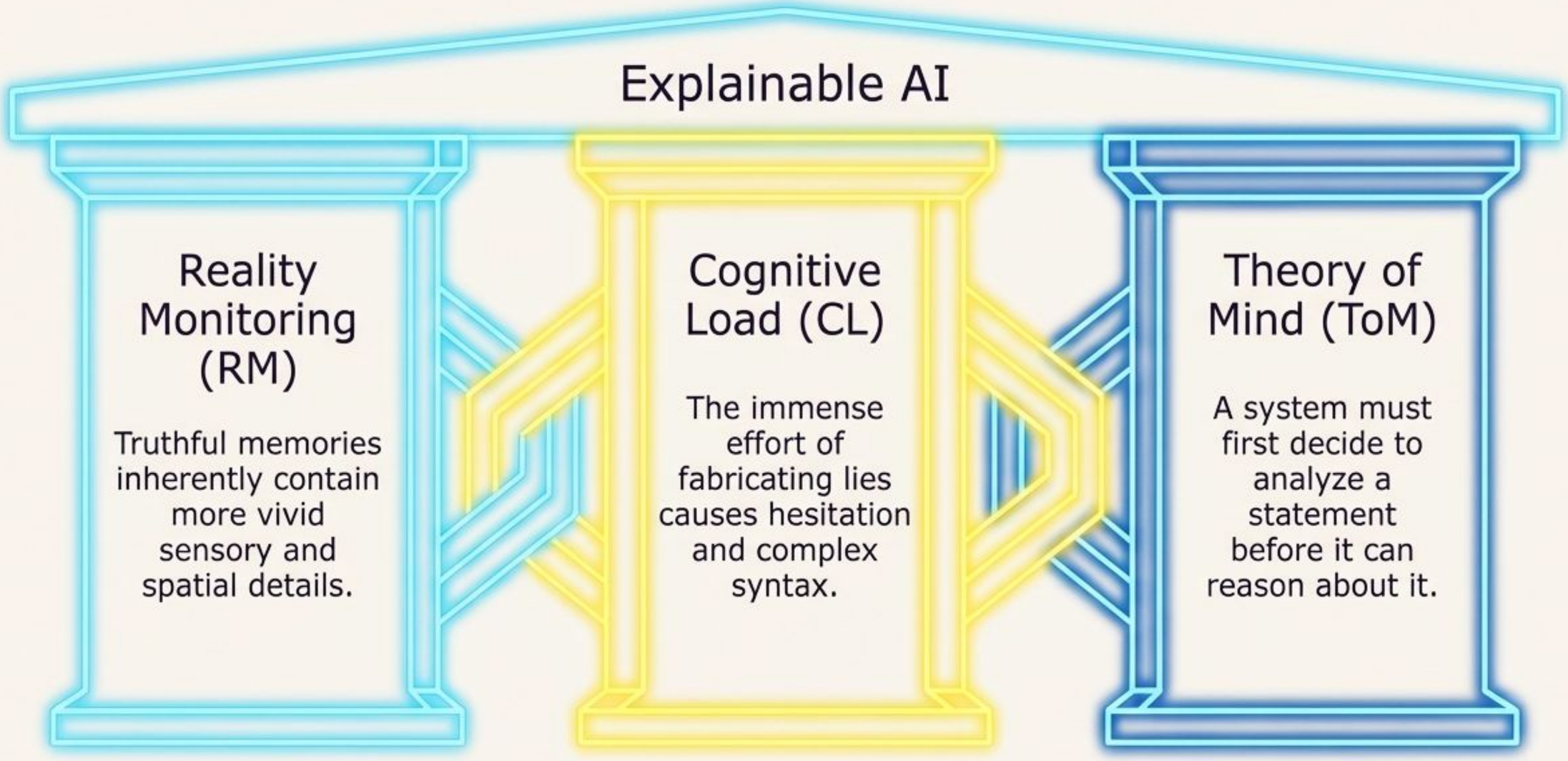
Automated models consistently outperform human deception detection (up to 77.3% vs. 54–59%).

However, these systems fail to explain why a statement is classified as deceptive.

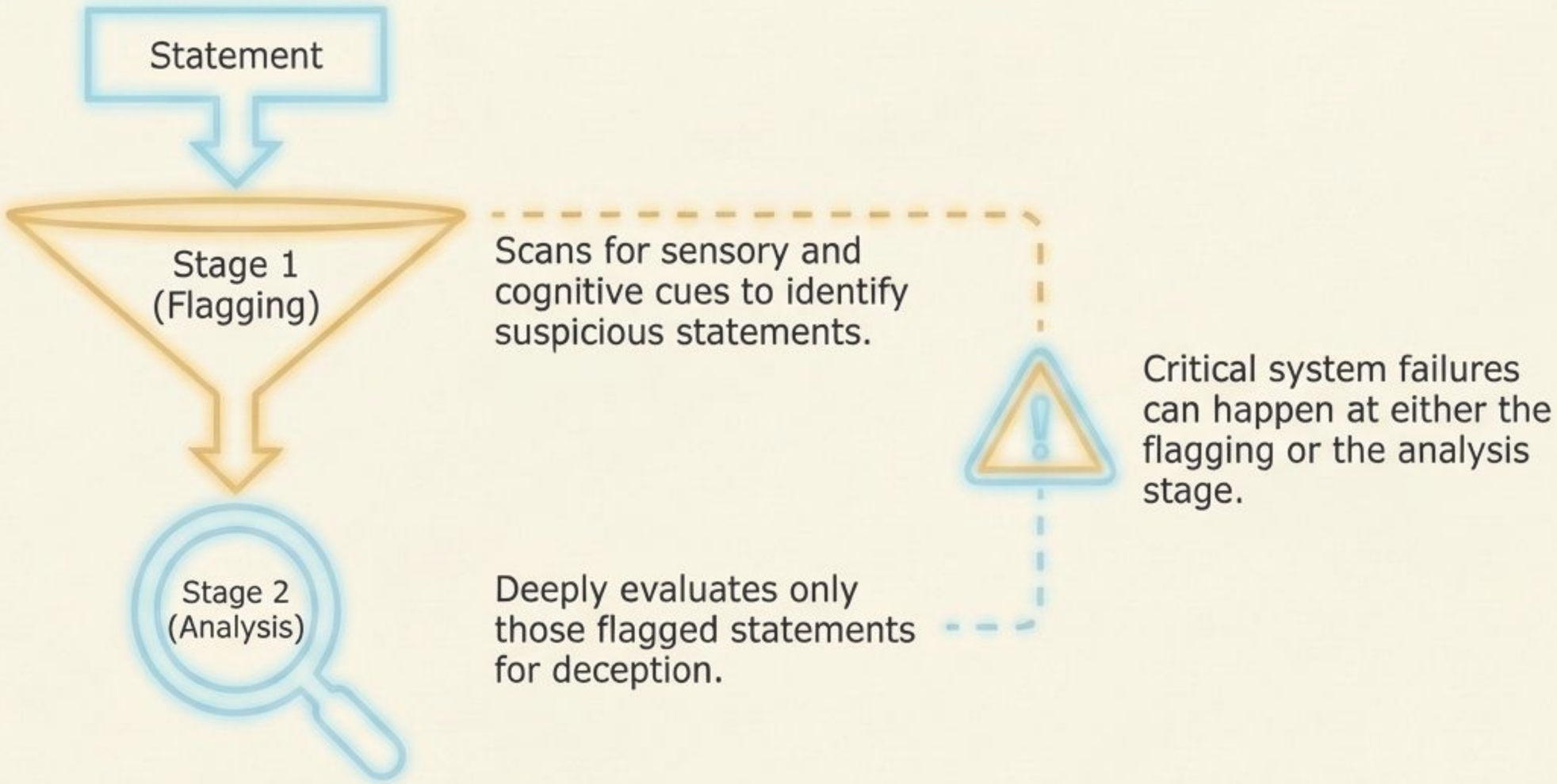
In high-stakes legal proceedings, unexplained misclassifications carry severe, life-altering consequences.

We require a framework to decode and categorize these system errors.

Bridging Psychology and Computational Linguistics



How the System Thinks: A Two-Stage Process



Categorizing Deception Detection Failures

Type A: Over-Suspicion

Truthful statements falsely flagged due to excessive model mentalization (False Positives at Stage 1).

Type B: Under-Suspicion

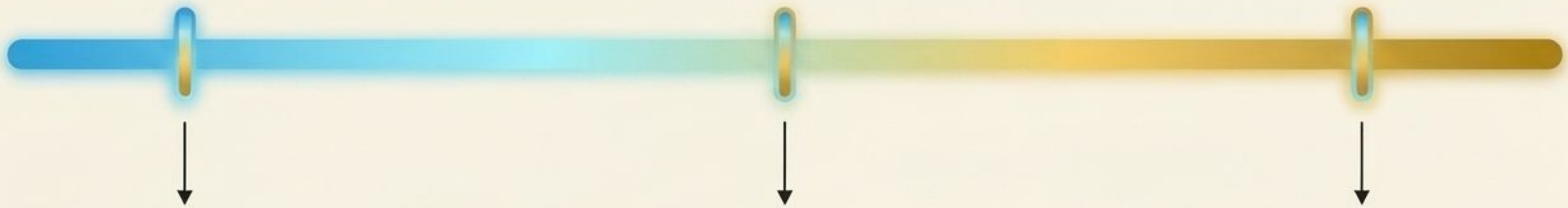
Deceptive statements that evade the flagging stage entirely (False Negatives at Stage 1).

Type C: Reasoning Failure

Correctly flagged statements that are ultimately misclassified during deep evaluation (Errors at Stage 2).

Quantifying Perception: The 0-5 Reality Monitoring Scale

1,190 sentences from the Trial dataset were coded for Visual, Auditory, and Spatiotemporal markers.



Score 0

Vague or absent sensory detail (e.g., I was there.)

Score 2

One specific detail (e.g., I saw a red car.)

Score 4

Highly specific detail (e.g., I saw a red car with a broken window.)

From Raw Testimony to Measurable Features

Data: 1,190 courtroom sentences

Reality Monitoring (0–5)

- Visual
- Auditory
- Spatio-temporal

Cognitive Load

- Hesitation
- Pauses
- Complexity

NLP Support (RoBERTa)

- Contextual patterns
- Semantic representation

Output: *Structured feature representation*

From Features to Error Types

Step 1: Flagging (Stage 1)

→ Is it suspicious?

Step 2: Analysis (Stage 2)

→ Is it deceptive?

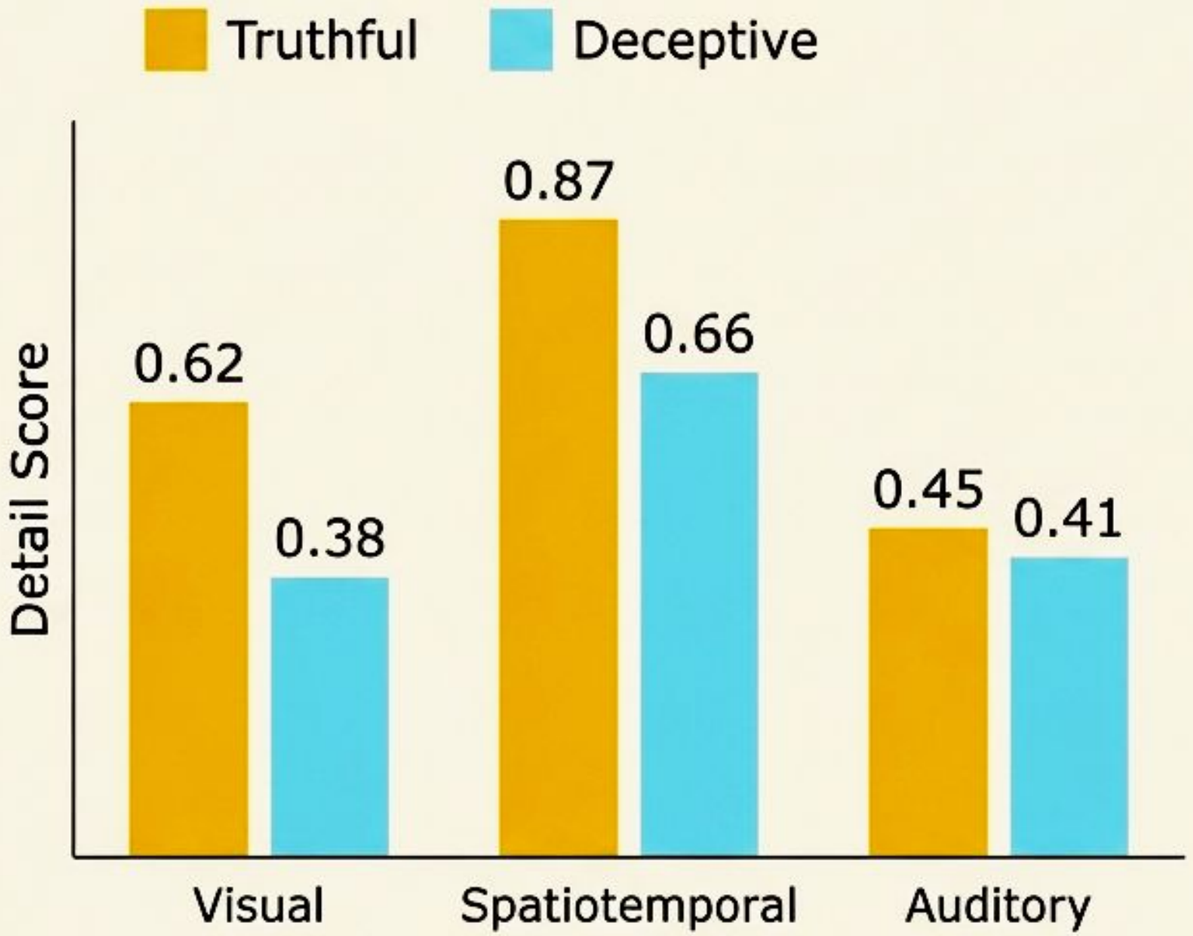
Error Mapping

- Type A: False Positive
- Type B: Missed Suspicion
- Type C: Misclassification

Input: *RM + CL + NLP features*

Output: *Error distribution & patterns*

Finding 1: The Sensory Deficit of Deception



- Truthful statements contain significantly more visual and spatiotemporal markers.
- Auditory details proved uniformly sparse across both categories, reflecting the strictly verbal nature of courtroom testimonies.
- Deceptive statements intentionally lack the perceptual anchors found in real memories.

Finding 2: The Dominance of Under-Suspicion

78.7%

- Nearly 80% of deceptive testimonies scored **extremely low** on Reality Monitoring details (≤1).
- This minimal detail allows lies to effortlessly evade the initial AI flagging stage **(Type B Error).**
- Conversely, only 3.3% of truthful testimonies contained enough detail to trigger false suspicion **(Type A Error)**.

Explaining the AI Failure in Court

The Human Cause

- Courtroom liars suffer from extreme cognitive load due to high emotional stakes.
- To manage this mental effort, they actively strip their narratives of complex perceptual details.

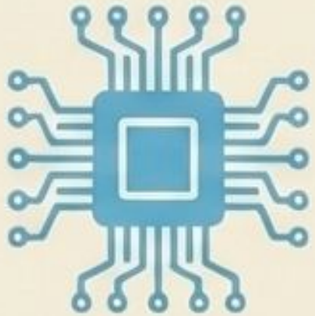
The Machine Effect

- Because standard AI heavily relies on sensory cues to flag deception, this detail stripping effectively cloaks the lie.
- Sensory analysis alone is catastrophically insufficient for high-stakes forensics.

Advancing the Field of Forensic Linguistics

Psychology

Proves that risk level acts as a strict moderator for how cognitive load affects language.



Technology

Demands that future systems combine sensory tracking with linguistic complexity metrics.

Law

Provides legal defense teams a transparent, peer-reviewed vocabulary to question AI reliability.

From Classifying Whether to Justifying Why

Automated systems must do more than classify text; they must **justify** their reasoning.

By mapping computational errors to human psychology, we create **accountability** in the legal system.

The dominance of **Type B errors** reveals a profound truth: the **ultimate courtroom deception is omission**, not **fabrication**.